Companion essay

Arithmetic Inherits the Geometry of its Basis

Metrics on the lattice, Pick's theorem, and the determinant as ruler

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Interactive companions: Pick learner · Lattice metrics · B²/A explorer

Abstract

A lattice is a discrete grid of points, but the notion of distance on a lattice is not unique. Four natural measures present themselves: the Euclidean metric inherited from the ambient plane, the Chebyshev metric of king moves on a chessboard, the Manhattan metric of taxi rides on a grid, and the lattice's own word length counted in primitive steps. Each captures something different. Pick's theorem turns out to be written in the last of these, and the determinant of the basis is the exchange rate that converts word length back into Euclidean area. This essay traces the thread. The square lattice is not a default but a special case, and the same machinery works on any lattice, including the triangular grid of the Eisenstein integers.

What is distance on a lattice?

A lattice is just a grid of points with whole-number coordinates: the integers $\mathbb{Z}^2$ in the plane, familiar from graph paper. But the moment we ask how far apart two points are, the question splits into several genuinely different ones.

A crow flying between $(0,0)$ and $(3,4)$ covers a Euclidean distance of $\sqrt{3^2+4^2}=5$. A chess king reaches the same point in $\max(3,4)=4$ moves, since diagonal and axis moves cost equally. A taxi confined to north-south and east-west streets needs $|3|+|4|=7$ blocks. And the lattice itself, asked how many of its own irreducible steps the journey consists of, replies one, because $(3,4)$ is a primitive vector with $\gcd(3,4)=1$ and so cannot be broken into smaller lattice pieces.

Metric	Generators	$d\big((0,0),(3,4)\big)$	$d\big((0,0),(6,8)\big)$	Character
Euclidean L^2	all directions in $\mathbb{R}^2$	5	10	ambient
Chebyshev L^∞	8 king-move directions	4	8	coarsens Euclidean
Manhattan L^1	4 axis directions	7	14	further coarsening
Lattice word length	all primitive vectors	1	2	intrinsic to $\mathbb{Z}^2$

The Euclidean, Chebyshev, and Manhattan metrics import their structure from the ambient plane. They restrict $\mathbb{R}^2$ geometry to a discrete subset, or distort it by treating certain directions as equally costly. The lattice word length is different in kind. It needs no reference to $\mathbb{R}^2$ at all, because it counts only what the lattice itself produces: how many irreducible generators a vector decomposes into.

Three metrics live in the ambient plane and restrict to the lattice. The fourth lives entirely on the lattice and ignores the plane.

Why gcd measures irreducibility

The lattice length of a segment, in its own primitive direction, is $\gcd(|a|,|b|)$. This formula asks a concrete question: how many lattice points lie strictly inside the segment from $(0,0)$ to (a,b)?

The points on the segment are parametrised as (ta, tb) for $t \in [0,1]$. Such a point is a lattice point exactly when both ta and tb are integers, which happens for $t = k/\gcd(a,b)$ with $k = 0, 1, \ldots, \gcd(a,b)$. The segment therefore contains $\gcd(|a|,|b|) + 1$ lattice points including endpoints, and $\gcd(|a|,|b|) - 1$ strictly interior ones.

A vector is *primitive* when $\gcd(|a|,|b|) = 1$, meaning the segment has *no* interior lattice points. The lattice cannot subdivide it further. The Pythagorean step $(3,4)$ is primitive: the segment from the origin to $(3,4)$ passes through no other lattice point, despite its Euclidean length of exactly 5. Its double $(6,8)$ has $\gcd = 2$: the midpoint $(3,4)$ now lies on the segment, splitting it into two primitive sub-steps. So $(6,8)$ has word length 2.

The word length is a count of primitive sub-steps. A vector (a,b) decomposes uniquely as $\gcd(a,b)$ copies of the primitive vector $(a/\gcd, b/\gcd)$. The lattice sees that decomposition and counts it. All Euclidean length information is discarded; the step $(1,0)$ and the step $(3,4)$ both cost 1 because both are primitive.

Chebyshev and Manhattan as coarsenings

Both Chebyshev and Manhattan are special cases of the word-length idea: hey are finite-generator word metrics, whereas B uses the primitive decomposition of each boundary edge.. Chebyshev uses the 8 shortest primitives (chess king moves), and Manhattan (chess rook moves) uses only the 4 axis vectors. Each step is still counted as cost 1, but only the chosen generators are allowed. Longer primitive steps that the full word length sees as single irreducible moves must be decomposed.

The result is that Chebyshev and Manhattan are coarser than the lattice word length. They collapse distinctions that the word length preserves, but they still inherit its combinatorial character. The Euclidean metric, by contrast, is finer than all three because it sees the actual continuous lengths of every edge.

Interactive companion: `lattice_metrics_v2.html`
(Click any lattice point to compare the four distances to the origin)

Pick's theorem is written in word length

A lattice polygon has vertices at lattice points. Walk along its boundary, edge by edge, and ask which metric counts the cost of that walk most naturally.

The classical Euclidean perimeter P measures each edge by its true length in $\mathbb{R}^2$. The Chebyshev perimeter would count max-steps. The Manhattan perimeter would count axis-only steps. But there is a fourth quantity that arises directly from the lattice, and it turns out to be the one Pick's theorem uses.

The boundary point count is a word length

Let B be the total number of lattice points lying on the boundary of the polygon. A single edge from vertex $\mathbf{u}$ to vertex $\mathbf{v}$ has displacement $(\Delta a, \Delta b) = \mathbf{v} - \mathbf{u}$. By the parametrisation argument above, this edge contains exactly $\gcd(|\Delta a|, |\Delta b|)$ primitive sub-steps, and therefore contributes $\gcd(|\Delta a|, |\Delta b|)$ to B.

Summing over all edges:

$$B = \sum_{\text{edges}} \gcd(|\Delta a_i|, |\Delta b_i|).$$

This is the lattice word length of the boundary cycle. Each primitive sub-step contributes one unit, regardless of its Euclidean length.

Pick's theorem in its native language

With B understood as a word-length count and I as the number of strictly interior lattice points, Pick's theorem reads:

$$A = I + \frac{B}{2} - 1$$

on the square lattice. Three lattice-intrinsic quantities (I, B, the constant 1) combine to give the Euclidean area A. The geometry of $\mathbb{R}^2$ enters only through A on the left; everything on the right is internal to $\mathbb{Z}^2$.

The proof, via Euler

Pick's formula is proved by recruiting the same Euler relation $V - E + F = 2$ that governs convex polyhedra. The setting moves down a dimension: instead of a polyhedron deformed onto a sphere, we have a polygon decomposed into triangles in the plane, with the result that the same combinatorial relation holds for any planar graph.

Subdivide the polygon into *primitive triangles*: triangles whose only lattice points are their three vertices. Let T be the number of triangles obtained. The vertices of the planar graph are the lattice points of the polygon: $V = I + B$. The faces are the T triangles plus the single unbounded outer face, so $F = T + 1$. Euler's relation gives

$$(I + B) - E + (T + 1) = 2, \qquad \text{so} \qquad T = E - I - B + 1.$$

Count edges two ways. Each interior edge of the triangulation is shared by two triangles, and each boundary edge of the polygon borders only one. Every triangle has three edges, so $3T = 2E_{\text{interior}} + E_{\text{boundary}} = 2E - B$, giving $E = (3T + B)/2$. Substituting back into the Euler expression:

$$T = \frac{3T + B}{2} - I - B + 1 \quad \Longrightarrow \quad T = 2I + B - 2.$$

The area formula now hinges on a single geometric fact about each primitive triangle. Each one has area exactly $\frac{1}{2}$, because its two edge vectors from any vertex form a basis for $\mathbb{Z}^2$ and the determinant of that basis has $|\det| = 1$. The total area is therefore

$$A = T \cdot \tfrac{1}{2} = \tfrac{1}{2}(2I + B - 2) = I + \frac{B}{2} - 1. \qquad \square$$

The pivotal fact: every primitive triangle on $\mathbb{Z}^2$ has area $\frac{1}{2}$ because the unimodular condition $|\det[\mathbf{v}_1, \mathbf{v}_2]| = 1$ makes it half a unit parallelogram. The Euler/triangulation argument is purely combinatorial; the determinant is the only geometric input.

A basis chooses directions. The determinant measures the area enclosed by those directions. Every lattice formula inherits that choice of area.

Interactive companion: `euler_pick_lattice_learner_v3.html`
(Drag a polygon, watch Pick verify $A = I + B/2 - 1$)

The determinant as a ruler

A lattice is generated by a choice of two basis vectors. Different choices give different fundamental parallelograms, but the same set of lattice points. The arithmetic of $\mathbb{Z}^2$ is unchanged; only the geometric ruler changes.

Three quantities suffice

The area of the fundamental parallelogram spanned by basis vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ depends on three things: the two lengths and the angle between them.

$$\Delta = |\mathbf{b}_1|\,|\mathbf{b}_2|\,\sin\theta.$$

This is the absolute value of the determinant of the basis matrix. The determinant is "angle-corrected product of lengths": when the basis is orthogonal the sine term disappears, and when both basis vectors have unit length and meet at right angles the determinant reduces to 1.

Basis type	**Lengths**	**Angle**	$\mathbf{\Delta}$
General	arbitrary	arbitrary	$\lvert\mathbf{b}_1\rvert\lvert\mathbf{b}_2\rvert\sin\theta$
Orthogonal	arbitrary	$90°$	$\lvert\mathbf{b}_1\rvert\lvert\mathbf{b}_2\rvert$
Orthonormal	unit	$90°$	1
Eisenstein	unit	$60°$	$\sqrt{3}/2$

The integer square lattice $\mathbb{Z}^2$ with its standard basis sits in the third row. Orthonormality is the special configuration in which the determinant happens to equal one.

Why the determinant disappears on the square lattice

On $\mathbb{Z}^2$ the classical Pick formula $A = I + B/2 - 1$ carries no prefactor. The determinant has not vanished from the geometry; it has become the unit by which all areas are measured. The determinant is hidden in the same way that multiplying by 1 is hidden.

The Gaussian integers $\mathbb{Z}[i]$ conceal Δ because their basis is orthonormal. The moment we choose a basis that is not orthonormal, the determinant becomes visible. The Eisenstein integers $\mathbb{Z}[\omega]$ provide the cleanest example.

The Eisenstein lattice

Let $\omega = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2} i$, a primitive cube root of unity. The *Eisenstein integers* $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$ form a triangular lattice in the complex plane, the densest packing of equal circles, with basis vectors 1 and ω meeting at 60°. The basis matrix is

$$M_\omega = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{bmatrix}, \qquad \det(M_\omega) = \frac{\sqrt{3}}{2}.$$

The Eisenstein basis matrix, viewed as a map from $\mathbb{Z}^2$ into $\mathbb{R}^2$, has determinant $\sqrt{3}/2 \neq 1$. The lattice is not unimodular relative to the Euclidean embedding.[1]

Two basis matrices side by side:

Square lattice $\mathbb{Z}^2$	**Eisenstein lattice $\mathbb{Z}[\omega]$**
$M_\square = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$M_\omega = \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} \end{bmatrix}$
$\det = 1 \cdot 1 - 0 \cdot 0 = \mathbf{1}$	$\det = 1 \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$
Basis at 90°. Unimodular.	*Basis at 60°.*
Δ hidden: equals the unit.	*Δ revealed: $\neq 1$.*
Primitive triangle area $= \frac{1}{2}$.	*Primitive triangle area $= \frac{\sqrt{3}}{4}$.*

The combinatorial scaffolding of Pick's proof (triangulate, apply Euler, count edges) does not care about which lattice it sits on. The only line that changes is the area per primitive triangle, which is now $\frac{1}{2}|\det(M_\omega)| = \frac{\sqrt{3}}{4}$. Pick's formula scales by exactly Δ:

$$\boxed{A = \Delta \cdot \left(I + \frac{B}{2} - 1\right), \qquad \Delta = \det(M_{\text{basis}}).}$$

[1]Within the Eisenstein lattice itself, the transition matrix between any two valid Eisenstein bases has determinant a unit of $\mathbb{Z}[\omega]$, that is ± 1 or $\pm\omega$ or $\pm\omega^2$. So the lattice is unimodular as an abstract $\mathbb{Z}$-module. What changes between lattices is the Euclidean area of the fundamental parallelogram, captured by $|\det(M_{\text{basis}})|$ in the ambient $\mathbb{R}^2$.

The boundary count B remains a lattice word length. The interior count I remains an integer. Only the conversion factor Δ from word length to Euclidean area changes. The square lattice result is the special case $\Delta = 1$.

One Pick formula serves every lattice; Δ is the single parameter that distinguishes them.

The cosine rule confirms the determinant

A satisfying elementary check. The determinant of M_ω equals $|\mathbf{f}_1||\mathbf{f}_2|\sin 60^\circ = \sqrt{3}/2$, the cross-product area formula. But the same answer arises from the cosine rule. The diagonal of the fundamental rhombus has length

$$d^2 = 1 + 1 - 2\cos 60^\circ = 1,$$

so $d = 1$. The rhombus splits into two isosceles triangles of area $\frac{1}{2}\sin 60^\circ = \frac{\sqrt{3}}{4}$, giving total area $\frac{\sqrt{3}}{2}$. The determinant and the cosine rule are measuring the same fundamental cell by different routes.

Boundary, area, and the question of shape

Once Pick's theorem is in place, a natural question follows: how should one measure how "efficiently" a polygon encloses area? The classical answer is the isoperimetric ratio P^2/A, which is minimised at 4π by the circle.

Translating this to the lattice raises a subtle point. The boundary count B is a word length, not a Euclidean perimeter. So the natural lattice analogue is B^2/A, with B in word-length units and A given by Pick. Substituting:

$$\frac{B^2}{A} = \frac{B^2}{\Delta(I + B/2 - 1)}.$$

The determinant sits in the denominator.

The scale-invariant combination is B^2/A, which transforms as $n^2B^2/n^2A = B^2/A$ is the right quantity to compare across shapes and sizes.

Under dilation of the polygon by an integer factor n, every coordinate scales by n. Each edge displacement $(\Delta a, \Delta b)$ becomes $(n\Delta a, n\Delta b)$ with $\gcd$ now multiplied by n. So $B \to nB$ and $A \to n^2 A$. The ratio $B/A \to 0$ for any polygon regardless of shape. B/A carries no shape information, only size.

What B^2/A captures and what it does not

B^2/A measures how efficiently a polygon encloses area, where boundary is counted in primitive lattice steps. It is comparable to the Euclidean ratio P^2/A but not identical to it, because B discards Euclidean length information. A diagonal step $(1,1)$ contributes 1 to B but $\sqrt{2}$ to P.

This means B^2/A can drop below 4π without violating any classical bound, because $B \leq P$ when oblique steps are present. The lattice ratio is a genuinely different quantity, comparable in spirit to P^2/A but measured in the lattice's own metric rather than in $\mathbb{R}^2$.

The hierarchy of boundary measures. For a fixed polygon:

Euclidean perimeter P	sum of edge lengths in $\mathbb{R}^2$
Word length B	sum of primitive sub-steps
Chebyshev word length	primitive sub-steps with only king-move generators
Manhattan word length	primitive sub-steps with only axis generators

Pick's theorem uses the second. $B \leq P$ in general, with equality only when every boundary edge is a unit vector. The choice of metric determines which inequality B^2/A obeys.

Interactive companion: `pa_law_v5.html`
(Draw polygons on either lattice, compare B/A and B^2/A under scaling)

Retracing the thread

The essay began with what seemed an elementary question: how do you measure distance on a lattice? Four answers emerged. Three of them (Euclidean, Chebyshev, Manhattan) restricted ambient geometry to a discrete subset. The fourth (lattice word length) used only the lattice itself, counting irreducible primitive steps with no reference to $\mathbb{R}^2$.

Pick's theorem turned out to be written in the fourth language. Its boundary count B is the word length of the boundary cycle, and its constant $\frac{1}{2}$ for primitive triangles arises from the unimodular determinant condition. Once those facts are recognised, the determinant Δ becomes visible as the conversion factor between word length and Euclidean area. The square lattice hides this because $\Delta = 1$. The Eisenstein lattice reveals it because $\Delta = \sqrt{3}/2$. Both follow the same Pick formula; only Δ differs.

$$A = \Delta \cdot \left(I + \tfrac{B}{2} - 1\right)$$

B is the word length of the boundary; Δ converts word length to Euclidean area; the basis is the ruler.

This is what it means to say that arithmetic inherits the geometry of its basis. The combinatorial structure of $\mathbb{Z}^2$ (or any lattice) gives rise to a natural word-length metric. Pick's theorem is the bridge between that metric and the ambient Euclidean area, and the determinant of the basis is the exchange rate.

Where this opens out

In the Gaussian and Eisenstein cases, the determinant is the Euclidean shadow of a deeper arithmetic invariant, the *field discriminant* of the underlying number field. The Eisenstein integers $\mathbb{Z}[\omega]$ form a unique factorisation domain; the class number of $\mathbb{Q}(\omega)$ is 1. The lattice determinant $\sqrt{3}/2$ is the geometric trace of that arithmetic fact.

The same machinery extends to higher-dimensional lattices arising from algebraic number fields. The Heegner numbers identify the imaginary quadratic fields of class number 1, an often-told story that connects unique factorisation to the remarkable near-integer $e^{\pi\sqrt{163}}$. The thread that began with counting how many primitive steps make up a lattice vector leads, eventually, to some of the deepest structures in number theory.

Along the way the determinant never changes its role. It measures the fundamental cell of the chosen basis, whether that basis belongs to the square lattice, the Eisenstein lattice, or the higher-dimensional lattices of algebraic number theory. The familiar Cartesian grid simply makes the determinant easy to overlook.